

6. (a) Explain principle of Ultrasonic machining process. Discuss its important process parameters. 7
- (b) Explain Principle of EDM non-conventional machining process with reference to Control circuits, MMR and surface finish. 8
7. (a) Sketch and explain screw moulding process of plastic. 7
- (b) Write the basic steps in Powder Metallurgy process. Explain the difference between infiltration and impregnation with suitable example. 8



Roll No.

Total Pages : 04

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B.Tech. (Mech.) (Sixth Semester)

Manufacturing Technology (PCC-ME-602/21)

Time : 3 Hours]

[Maximum Marks : 75

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Define terms : Reaming, Boring and Broaching. 1.5
- (b) Define term 'fool proofing'. 1.5
- (c) Define term 'Spring back'. 1.5
- (d) Give reason for preference of high strength to weight ratio of forged part. 1.5
- (e) State Taylor's Principle for Gauge design. 1.5

- (f) What is zero line for Limit systems ? 1.5
- (g) Differentiate between standard gauge and a limit gauge. 1.5
- (h) State the principle of operation of a comparator. 1.5
- (i) Write the Principle of Laser beam machining (LBM). 1.5
- (j) Define terms : Pre-coated Powder and Pre-alloyed Powder. 1.5

Part B

- 2. (a) How Rotation and lifting of a drill bushing be prevented ? 7
- (b) Discuss the different types of boring machines with neat sketches. 8
- 3. (a) Explain the method to reducing cutting forces in Punch. 7
- (b) Sketch a sectional view of a blanking die, and label on it- Clearance, Straight and angular clearance. 8

- 4. (a) Explain and compare 'Hole basic system' and 'Shaft basic system' for fits. 7
- (b) A bore of LL-27.50, HL-27.52 mm diameter and 45 mm long, is to be checked. Design, draw and dimension a Plug gauge for this based on Taylor's principle of gauge design. 8

- 5. (a) Calculate the fundamental deviations and tolerances and hence obtain the limits of sizes for hole and shaft in the fit : $45 H_8/f_8$. The diameter of the steps are 30 mm and 50 mm. The fundamental deviation for d shaft is given as $-5.5D^{0.41}$. the tolerance unit is:

$$i = 0.45\sqrt[8]{D} + 0.001D$$

The tolerance grade for number 8 quality is 25i. 10

- (b) Explain Principle of Sine Bar. Also write its applicability. 5